

Project Milestone: Multi-Label Toxic Comment Error Analysis

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2. Problem Description

This project investigates multi-label toxic comment classification using the Kaggle Jigsaw dataset. Our goal is not only to build toxic comment classifiers but also to analyze why they fail, especially when comments contain overlapping toxicity categories and when rare labels such as threat, severe_toxic, and identity_hate are underrepresented.

3. Data

We use the [Kaggle Jigsaw Toxic Comment Classification](#) train.csv, with 159,571 Wikipedia comments and six non-exclusive labels: toxic, severe_toxic, obscene, threat, insult, and identity_hate. Since comments may receive multiple labels, this is a supervised multi-label task. Label counts show strong imbalance, especially for threat, severe_toxic, and identity_hate. We checked missing values, duplicate IDs/comments, binary label validity, and corrupted entries.

4. Baseline

Our baseline is a TF-IDF + One-vs-Rest Logistic Regression classifier. Each comment is converted into TF-IDF features using unigrams and bigrams, and each toxicity category is treated as a separate binary classification task. This allows one comment to receive multiple labels. We use the default probability threshold of 0.5 to convert predicted probabilities into binary label predictions. Baseline is intentionally simple and interpretable to provide a reliable starting point for later threshold tuning and structured error analysis.

5. Evaluation Methodology

We evaluate the baseline using an 80/20 train-validation split with random seed 42. Because this is a highly imbalanced multi-label task, we report micro-F1, macro-F1, weighted-F1, Hamming loss, and per-label precision, recall, and F1-score. We focus especially on macro-F1 and rare-label recall. We also save validation-level predictions and probabilities to support later error analysis, including false positives, false negatives, single-label versus multi-label errors, and partial misses.

6. Results – Detailed output tables are included in Appendix A & GitHub Link

The baseline achieved a micro-F1 of 0.661, macro-F1 of 0.457, weighted-F1 of 0.651, and Hamming loss of 0.0198. It performed better on frequent labels such as toxic, obscene, and insult, but much worse on rare labels such as severe_toxic, identity_hate, and threat. The threat label had low recall, showing that the model misses most true threat comments.

These milestone results confirm that the baseline is useful not only as an initial classifier but also as a diagnostic tool for identifying systematic error patterns. In the next stage, we will extend our label co-occurrence analysis, compare single-label versus multi-label comments, identify partial misses, and test imbalance mitigation methods such as label-specific threshold tuning and class-weighted Logistic Regression. We will also add ROC-AUC as an additional probability-based evaluation metric and compare whether a BERT-based transformer model can reduce the same error patterns.

Appendix

Appendix A. Baseline Output Tables

Overall metrics table:

metric	value
micro_f1	0.6609579876906609
macro_f1	0.45679065319745477
weighted_f1	0.651148468622832
hamming_loss	0.019849600501331662

Per-label metrics table:

label	precision	recall	f1	true_positive	false_positive
toxic	0.92125579001544	0.5857329842931938	0.7161432286457291	1790	153
severe_toxic	0.5964912280701754	0.2118380062305296	0.31264367816091954	68	46
obscene	0.9276437847866419	0.5830903790087464	0.7160759040458289	1000	78
threat	0.8571428571428571	0.08108108108108109	0.14814814814814814	6	1
insult	0.8515376458112407	0.49752168525402723	0.6280797809933516	803	140
identity_hate	0.7307692307692307	0.1292517006802721	0.21965317919075145	38	14

false_negative	true_negative	false_positive_rate	false_negative_rate	positive_count_in_validation
1266	28706	0.005301639003430472	0.4142670157068063	3056
253	31548	0.0014559726530353864	0.7881619937694704	321
715	30122	0.0025827814569536426	0.41690962099125367	1715
68	31840	3.140604880499984e-05	0.918918918918919	74
811	30161	0.00462030956074057	0.5024783147459727	1614
256	31607	0.0004427437462445843	0.8707482993197279	294

Project GitHub Link:

<https://github.com/JunhaoLiu3/MSE641-Project-Toxic-Comment-Classification>